

Mathematics for Machine Learning

Part II: Vectors, Matrices and Linear Algebra

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Roadmap

- 1 Chapter 1: Vectors
- 2 Chapter 2: Matrices
- 3 Wrapping Up

Picking Up Where Part I Left Off

Three Promises We Now Keep

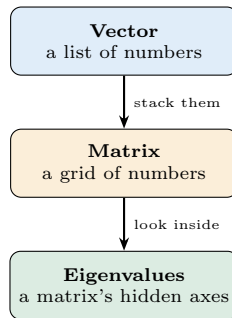
What You Already Know

In Part I the **gradient** was “an ordered list of numbers” — one slope per parameter. That list was a **vector** all along. This deck gives vectors and their bigger cousins, matrices, their full toolkit.

The Three Loose Ends

- 1 The total differential was written as a long **sum**. It is really a **dot product**.
- 2 The Hessian was shown as a **table**. It is really a **matrix**.
- 3 Classifying critical points was postponed because it needs **eigenvalues**.

All three are settled in this deck.



One data point → a whole dataset → its underlying structure.

Chapter 1: Vectors

1.1 What Is a Vector?

An Ordered List of Numbers

Definition

A **vector** is an ordered list of numbers. In code it is a 1-D array. We write it as a bold lowercase letter:

$$\mathbf{x} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \in \mathbb{R}^2$$

Ordered matters: $[2, 3]$ and $[3, 2]$ are different vectors.

Numerical Example

A house of 1200 sqft with 3 bedrooms becomes a **feature vector**:

$$\mathbf{h} = \begin{bmatrix} 1200 \\ 3 \end{bmatrix}$$

Machine Learning Connection

A vector is one **data point**: a customer, a house, a row of pixels, or a word embedding.

The gradient from Part I was also a vector — one slope per parameter.

Two Ways to See It

- **As a list**: data to compute with.
- **As an arrow**: a direction and a length in space.

1.2 Dimensions and Plotting

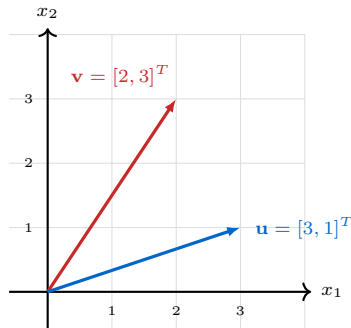
Arrows From the Origin

Dimension = Number of Entries

- $\mathbf{v} = [2, 3]^T$ lives in $\mathbb{R}^2$
- $\mathbf{w} = [1, 5, -2]^T$ lives in $\mathbb{R}^3$
- A 784-pixel image lives in $\mathbb{R}^{784}$

Drawing Convention

We draw a vector as an arrow from the **origin** to the point given by its entries. Only the **direction** and **length** matter — the same arrow moved elsewhere is the same vector.



1.3 Vector Addition

Element-wise, and the Parallelogram Law

Algebraically: Add Entry by Entry

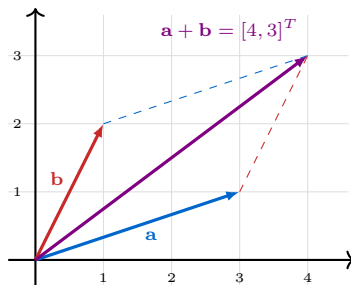
$$\mathbf{a} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\mathbf{a} + \mathbf{b} = \begin{bmatrix} 3 + 1 \\ 1 + 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

Geometrically: Two Readings

- **Tip-to-tail:** walk along $\mathbf{a}$, then along $\mathbf{b}$.
- **Parallelogram:** the sum is the diagonal of the parallelogram built on $\mathbf{a}$ and $\mathbf{b}$.

Both give the same arrow.



The dashed sides close the parallelogram; the purple diagonal is the sum.

1.4 Negation and Subtraction

Flipping Direction

Negative of a Vector

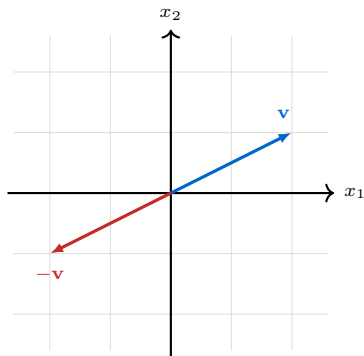
Negating flips the arrow to point the opposite way, keeping its length:

$$\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \implies -\mathbf{v} = \begin{bmatrix} -2 \\ -1 \end{bmatrix}$$

Subtraction Is Addition of the Negative

$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$$

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$



Same length, opposite direction — exactly 180° apart.

1.5 Subtraction: The Geometric Picture

The Arrow Between Two Tips

The Rule

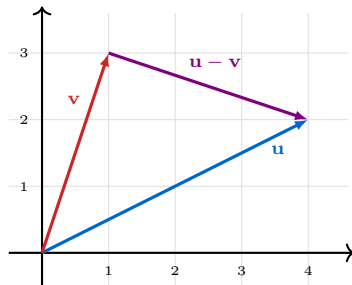
$\mathbf{u} - \mathbf{v}$ is the arrow drawn from the **tip of $\mathbf{v}$** to the **tip of $\mathbf{u}$** .

$$\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \Rightarrow \mathbf{u} - \mathbf{v} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

Memory aid: it points *towards* the first vector named.

Machine Learning Connection

$\mathbf{u} - \mathbf{v}$ is the **difference between two data points**. Its length $\|\mathbf{u} - \mathbf{v}\|$ is the **distance** between them — the basis of k -nearest-neighbours and of clustering.



The purple arrow does not start at the origin — but slide it back and its entries are $[3, -1]^T$.

1.6 Scalar Multiplication and Unit Vectors

Stretching Without Turning

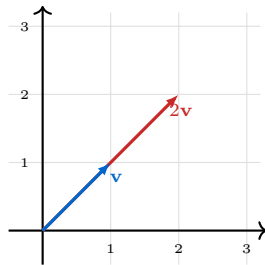
Scalar Multiplication

Multiplying by a number c **stretches or shrinks** the arrow but never turns it (a negative c flips it): $2 \cdot [1, 1]^T = [2, 2]^T$. Note $\|c \mathbf{v}\| = |c| \|\mathbf{v}\|$.

Unit Vector: Direction Only

A **unit vector** has length 1: divide a vector by its own length, $\hat{\mathbf{u}} = \mathbf{u} / \|\mathbf{u}\|$.

E.g. $\mathbf{u} = [3, 4]^T$ has $\|\mathbf{u}\| = 5$, so $\hat{\mathbf{u}} = [0.6, 0.8]^T$.



ML note: normalising to unit length is routine before comparing items, so **direction** is compared rather than size.

1.7 Magnitude: Norms

How Long Is a Vector?

L_2 Norm — Euclidean, Straight-Line

The everyday notion of length, straight from Pythagoras:

$$\|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

Example: $\mathbf{x} = [3, 4]^T \Rightarrow \sqrt{9 + 16} = 5$.

Other Norms

- L_1 (**Manhattan**): $|x_1| + |x_2| + \dots$ Example: $3 + 4 = 7$.
- L_p (**general**): $(\sum_i |x_i|^p)^{1/p}$.
- L_∞ : the single largest $|x_i|$. Here, 4.

Machine Learning Connection

Norms drive **regularization**, which shrinks model weights to prevent overfitting:

- An L_2 **penalty** on $\|\mathbf{w}\|_2^2$ shrinks all weights smoothly toward zero.
- An L_1 **penalty** on $\|\mathbf{w}\|_1$ drives some weights **exactly** to zero, switching those features off.

1.8 L_1 vs L_2 : The Geometry

Two Ways to Measure the Same Trip

For $\mathbf{x} = [3, 4]^T$

L_2 — as the crow flies:

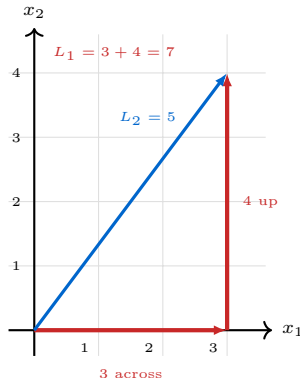
$$\|\mathbf{x}\|_2 = \sqrt{3^2 + 4^2} = 5$$

L_1 — along the city blocks:

$$\|\mathbf{x}\|_1 = |3| + |4| = 7$$

Why They Differ

L_1 must travel along the axes; L_2 may cut the corner. So $\|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$ always.



Red follows the streets; blue flies over the buildings.

1.9 The Dot Product

Two Definitions, One Number

Algebraic: Multiply and Add

$$\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i = a_1 b_1 + a_2 b_2 + \dots$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 4 \end{bmatrix} = (1)(3) + (2)(4) = 11$$

The output is a single **number**, not a vector.

Geometric: Length Times Alignment

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$$

Reading the Sign

- > 0 : the vectors broadly **agree** ($\theta < 90^\circ$)
- $= 0$: they are **perpendicular**
- < 0 : they broadly **oppose** ($\theta > 90^\circ$)

A Useful Special Case

Taking the dot product of a vector with itself gives its squared length:

$$\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2$$

which is why $\mathbf{a} \cdot \mathbf{a}$ can never be negative.

1.10 Cashing In a Promise From Part I

The Total Differential Is a Dot Product

What We Wrote in Part I

We estimated the change in a multivariable function as a long sum — sensitivity times movement, added over all inputs:

$$\Delta f \approx \frac{\partial f}{\partial x_1} \Delta x_1 + \cdots + \frac{\partial f}{\partial x_n} \Delta x_n$$

What It Really Was

That is precisely “multiply matching entries and add” — a dot product of the gradient with the step:

$$\Delta f \approx \nabla f \cdot \Delta \mathbf{x}$$

Same arithmetic, one symbol instead of n terms.

Machine Learning Connection

Using $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$, the change Δf is largest when $\Delta \mathbf{x}$ points **along** ∇f ($\cos \theta = 1$).

This is the **proof** of the Part I claim that the gradient points in the direction of steepest ascent — and that stepping along $-\nabla f$ decreases f fastest.

A prediction $\hat{y} = \mathbf{w} \cdot \mathbf{x} + b$ is also a dot product: weights against features.

1.11 Projection

How Much of One Vector Lies Along Another

The Formulas

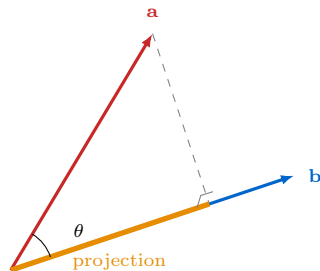
The **scalar projection** of **a** onto **b** — the length of the shadow:

$$\text{comp}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|}$$

The **vector projection** — the shadow itself, as an arrow along **b**:

$$\text{proj}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|^2} \mathbf{b}$$

Mental image: shine a light down onto **b**; the shadow **a** casts is the projection.



The dashed line meets **b** at a right angle. If $\mathbf{a} \perp \mathbf{b}$ the shadow has zero length.

1.12 Cosine Similarity

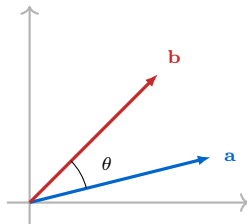
Comparing Direction, Ignoring Size

Definition

Rearranging the geometric dot product isolates the angle:

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$$

Dividing by both lengths cancels magnitude, leaving pure **orientation**.



Reading the Value

- $\cos \theta = 1$: same direction (most similar)
- $\cos \theta = 0$: orthogonal, unrelated
- $\cos \theta = -1$: exactly opposite

Machine Learning Connection

Cosine similarity is the standard measure for comparing **documents**, **word embeddings** and **users**. A long document and a short one on the same topic point the same way, even though their lengths differ wildly.

1.12 Cosine Similarity: Worked Example

Documents as Vectors

Numerical Example

Count two keywords, (*goal*, *election*), in two documents:

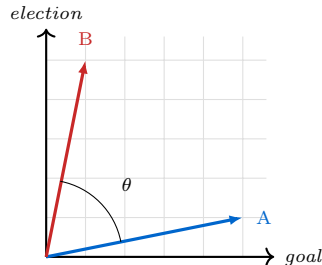
- Doc A (sports): $\mathbf{a} = [5, 1]^T$
- Doc B (politics): $\mathbf{b} = [1, 5]^T$

$$\mathbf{a} \cdot \mathbf{b} = (5)(1) + (1)(5) = 10$$

$$\|\mathbf{a}\| = \|\mathbf{b}\| = \sqrt{26} \approx 5.10$$

$$\cos \theta = \frac{10}{26} \approx \mathbf{0.38}$$

Low similarity — the documents point in noticeably different directions ($\theta \approx 67^\circ$).



Doubling every word count in Doc A would leave the **angle** — and so the similarity — completely unchanged.

1.13 Orthogonality

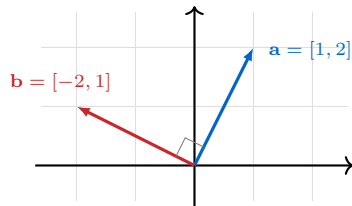
The Right-Angle Case, and Why It Is Special

Definition

Two non-zero vectors are **orthogonal** (perpendicular) exactly when their dot product vanishes:

$$\mathbf{a} \cdot \mathbf{b} = 0 \iff \mathbf{a} \perp \mathbf{b}$$

Example: $[1, 2] \cdot [-2, 1] = -2 + 2 = 0$. ✓



Orthonormal = Orthogonal + Unit Length

The standard axes $\hat{\mathbf{e}}_1 = [1, 0]^T$ and $\hat{\mathbf{e}}_2 = [0, 1]^T$ each have length 1 and meet at 90° .

Machine Learning Connection

Orthogonal features carry **non-overlapping information** — no redundancy, so each can be interpreted on its own.

1.14 Linear Combinations and Span

What Can You Build From a Set of Vectors?

Linear Combination

Scale some vectors and add the results:

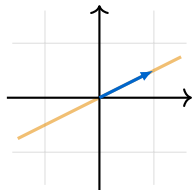
$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k$$

Example: $2[1, 0] + 3[0, 1] = [2, 3]$.

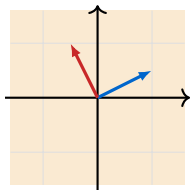
Span

The **span** of a set is every point you can reach by such combinations.

- One non-zero vector spans a **line**.
- Two independent vectors span a **plane**.
- Two vectors on the same line still span only a **line**.



span of **one** vector
a **line**



span of **two** independent
the whole **plane**

1.15 Linear Independence

Is Anyone Redundant?

The Test

Vectors are **linearly independent** if the only combination giving the zero vector uses all-zero coefficients:

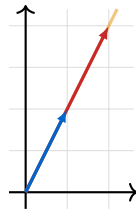
$$c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k = \mathbf{0} \implies c_1 = \cdots = c_k = 0$$

Otherwise they are **dependent**: one is a combination of the others and adds nothing new.

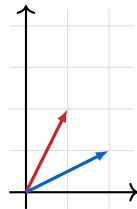
Numerical Example

$[1, 2]$ and $[2, 4]$ are **dependent**: $2[1, 2] - [2, 4] = \mathbf{0}$.

$[1, 2]$ and $[2, 1]$ are **independent** — neither is a multiple of the other.



dependent
same line



independent
different directions

1.16 Basis and Dimension

The Minimum Set That Builds Everything

Basis

A **basis** for a space is a set of vectors that is

- ① **independent** (nothing redundant), and
- ② **spanning** (reaches everything).

It is the smallest complete set of building blocks.
The **dimension** is how many vectors it contains.

The Standard Basis of $\mathbb{R}^2$

$$\hat{\mathbf{e}}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{e}}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 \\ 3 \end{bmatrix} = 2\hat{\mathbf{e}}_1 + 3\hat{\mathbf{e}}_2$$

Reading off coordinates is already using a basis — usually without noticing.

Machine Learning Connection

A basis **is not unique**. The same data can be described in many coordinate systems, and some are far more useful than others.

Choosing a **better basis** is a recurring move: describe the same data in coordinates lined up with its own structure, and hard problems often turn into easy ones.

Chapter 1 Take-Away: Vectors

- ➊ A **vector** is an ordered list of numbers — and also an arrow with a direction and a length. Both views are useful.
- ➋ **Norms** measure size. L_2 flies straight, L_1 walks the grid — and an L_1 penalty drives weights exactly to zero.
- ➌ The **dot product** multiplies matching entries and adds. Its sign says whether two vectors agree, oppose, or are perpendicular.
- ➍ $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$ gives **cosine similarity**, which compares direction while ignoring magnitude.
- ➎ **Dot product** = 0 means **orthogonal** — no shared information.
- ➏ **Span** is what you can build; **independence** asks whether anyone is redundant; a **basis** is the smallest set that builds everything.

Chapter 1 Quiz

Test Your Skills

- ❶ Let $\mathbf{u} = [2, -2]$. What is the unit vector $\hat{\mathbf{u}}$?
- ❷ If $\|\mathbf{a}\|_2 = 3$, $\|\mathbf{b}\|_2 = 4$, and the angle between them is 90° , what is $\mathbf{a} \cdot \mathbf{b}$?
- ❸ Calculate the L_1 norm of $\mathbf{v} = [-5, 10, -2]$.
- ❹ True or false: $\mathbf{a} \cdot \mathbf{a}$ can be negative.
- ❺ Given $\mathbf{x} = [1, 2]$ and $\mathbf{y} = [2, 4]$, find the cosine similarity without a calculator.
- ❻ If $\mathbf{a} \cdot \mathbf{b} = 0$, what does this imply geometrically?
- ❼ Are $[1, 3]$ and $[-2, -6]$ linearly independent?
- ❽ What is the span of $\{[1, 0], [2, 0]\}$?

Answers are in the **Appendix** at the end of the deck.

Chapter 2: Matrices

2.1 What Is a Matrix?

A Grid of Numbers — and a Stack of Vectors

Definition

A **matrix** is a rectangular grid of numbers:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

This is 2×3 — **rows first, columns second**.
The entry in row i , column j is a_{ij} ; here $a_{23} = 6$.

Two Readings

A **stack of row vectors** (one per sample), or a **set of column vectors** (one per feature). Both are used constantly.

Machine Learning Connection

In a dataset matrix X :

- **Rows** = samples (users, houses, images)
- **Columns** = features (age, income, height)

A 1000×20 matrix means 1000 examples described by 20 features each.

2.2 The Dataset View

From Table to Matrix

Sq. Feet x_1	Bedrooms x_2	Age x_3
2100	3	20
1400	2	15
2500	4	5

$$X = \begin{bmatrix} 2100 & 3 & 20 \\ 1400 & 2 & 15 \\ 2500 & 4 & 5 \end{bmatrix} \in \mathbb{R}^{3 \times 3}$$

Reading X

- A **row** is one house.
- A **column** is one feature measured across all houses.

A Practical Warning

Notice that column 1 runs in the thousands while column 2 is single digits. Most ML algorithms are sensitive to this mismatch, which is why features are usually **scaled** before training.

2.3 Special Types of Matrices

The Ones Worth Recognising

Square, Diagonal, Identity

- **Square:** rows = columns ($n \times n$). Only square matrices have determinants, eigenvalues or inverses.
- **Diagonal:** non-zeros only on the main diagonal.
- **Identity I :** a diagonal of 1s. It is the “1” of matrix multiplication: $AI = IA = A$.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix} \quad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Symmetric: $A = A^T$

A mirror image across the main diagonal — $a_{ij} = a_{ji}$:

$$A = \begin{bmatrix} 1 & 7 & 3 \\ 7 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

Matching colours mark the mirrored pairs.

Machine Learning Connection

Covariance matrices are always symmetric — how feature i varies with j is the same as j with i . The **Hessian** from Part I is symmetric too.

2.4 Transpose, Addition and the Frobenius Norm

Basic Operations

Transpose A^T

Flip rows into columns. An $m \times n$ becomes $n \times m$:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Rules: $(A^T)^T = A$, and $(AB)^T = B^T A^T$ — **order reverses**.

Addition

Element-wise, and only for **identical**

dimensions: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 3 & 5 \end{bmatrix}$

Frobenius Norm

The size of a whole matrix — square everything, add, take the root:

$$\|A\|_F = \sqrt{\sum_i \sum_j a_{ij}^2}$$

Numerical Example

For $A = \begin{bmatrix} 3 & -1 \\ 2 & 5 \end{bmatrix}$:

$$\|A\|_F = \sqrt{9 + 1 + 4 + 25} = \sqrt{39} \approx 6.24$$

It is the L_2 norm of the flattened matrix.

2.5 Two Different “Multiplications”

Do Not Confuse Them

1. Hadamard Product $A \odot B$

Element-wise; same shape in, same shape out.

This is `A * B` in NumPy.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \odot \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 3 & 8 \end{bmatrix}$$

2. Matrix Product $A \times B$

Entry (i, j) of the result is the **dot product** of row i of A with column j of B .

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

Written `A @ B`.

Order Matters

In general $AB \neq BA$. Multiplication is **not commutative**: rotating then stretching differs from stretching then rotating.

But These Still Hold

Associative: $(AB)C = A(BC)$. Distributive: $A(B + C) = AB + AC$.

You may **regroup**, but never **reorder**.

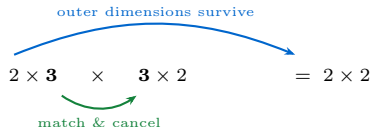
2.6 Conformability

Which Shapes Can Be Multiplied?

The Shape Rule

$$(m \times \mathbf{n}) \times (\mathbf{n} \times p) = (m \times p)$$

The **inner** dimensions must match and then cancel; the **outer** ones give the result's shape.



Numerical Example

A 2×3 times a 3×2 gives 2×2 :

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 7 & 8 \\ 9 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 31 & 19 \\ 85 & 55 \end{bmatrix}$$

Row 1 · Col 1 = $(1)(7) + (2)(9) + (3)(2) = 31$.

Swapping Changes Everything

Reversing the order above gives

$(3 \times 2)(2 \times 3) = 3 \times 3$ — a different **size**, not merely different numbers. Shape errors are the most common bug in ML code.

2.7 Matrix–Vector Multiplication

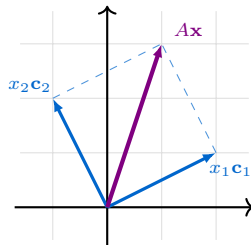
A Matrix Is a Function That Moves Vectors

As a Linear Combination of Columns

Writing A as its columns $\mathbf{c}_1, \mathbf{c}_2$:

$$A\mathbf{x} = [\mathbf{c}_1 \quad \mathbf{c}_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1\mathbf{c}_1 + x_2\mathbf{c}_2$$

So $A\mathbf{x}$ is always a **linear combination of A 's columns** — it can only ever land inside their span.



The Transformation View

Feed a vector in, get a vector out. The matrix **rotates, stretches, shears or squashes** the space.

The output is the parallelogram sum of the scaled columns.

2.7 Matrix–Vector Multiplication: Rotation

A Concrete Transformation

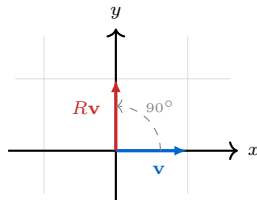
A 90° Counter-Clockwise Rotation

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Applied to the unit x -vector:

$$R \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

The x -axis has been carried onto the y -axis.



Numerical Example

On a general vector: $R[2, 3]^T = [-3, 2]^T$. Length is preserved ($\sqrt{13}$ both sides) — rotations never stretch.

Machine Learning Connection

Rotation matrices are **orthogonal**: $R^T R = I$, so $R^{-1} = R^T$. Inverting is free — just transpose. Any change to a new set of perpendicular axes is a matrix of this kind.

2.8 Rank

How Much Genuine Information Does a Matrix Carry?

Definition

The **rank** of a matrix is the number of **linearly independent** columns (equivalently, rows).

It is the true dimension of the space the columns can reach — how many directions the matrix genuinely spans.

Numerical Example

In $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, column 2 is $2 \times$ column 1, so $\text{rank}(A) = 1$: the matrix squashes the plane onto a line.

But $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ has $\text{rank} = 2$ — **full rank**.

Full Rank vs Rank-Deficient

For an $n \times n$ matrix:

- $\text{rank} = n$: **full rank**, $\det \neq 0$, inverse exists.
- $\text{rank} < n$: **rank-deficient** (singular), $\det = 0$, no inverse.

Machine Learning Connection

Rank deficiency means **redundant features**. If “height in cm” and “height in inches” are both columns, one is a multiple of the other: the rank drops, the matrix turns singular, and any method needing an inverse fails outright.

2.9 The Determinant

How Much Does a Matrix Scale Area?

For a 2×2 Matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \implies \det(A) = ad - bc$$

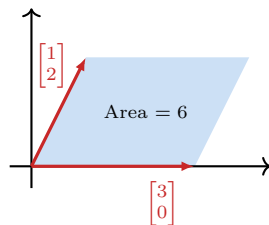
Example: $A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} \Rightarrow \det = 12 - 2 = 10.$

Why It Matters

- $\det(A) = 0 \Rightarrow A$ is **singular**: it collapses space onto a line or a point, and A^{-1} does **not** exist.
- A **negative** determinant means the transformation also flips orientation (a mirror image).

Geometric Meaning

$|\det(A)|$ is the **area of the parallelogram** built on A 's column vectors.



For $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$, $\det = (3)(2) - (1)(0) = 6. \checkmark$

2.9 Determinant: Minors and Cofactors

Building Blocks for Larger Matrices

Step 1 — The Minor M_{ij}

Delete row i and column j , then take the determinant of what remains. For

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{bmatrix}$$

$$M_{11} = \det \begin{bmatrix} 4 & 5 \\ 0 & 6 \end{bmatrix} = 24$$

Step 2 — The Cofactor C_{ij}

Attach an alternating sign: $C_{ij} = (-1)^{i+j} M_{ij}$.
Since $1 + 1$ is even, $C_{11} = +24$.

The Sign Checkerboard

The $(-1)^{i+j}$ factor is easiest remembered as a pattern:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Always start with $+$ in the top-left corner and alternate.

This machinery drives both 3×3 determinants and inverses.

2.9 Determinant of a 3×3 Matrix

Expansion Along the First Row

The Formula

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

Walk across the top row: entry $\times$ its cofactor, summed.

Practical Tip

Expand along whichever row or column has the **most zeros** — every zero kills an entire term. For a triangular matrix this reduces to multiplying the diagonal.

Numerical Example

For $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{bmatrix}$:

- $a_{11} = 1$, $M_{11} = \det \begin{bmatrix} 4 & 5 \\ 0 & 6 \end{bmatrix} = 24$, sign $+$ $\Rightarrow +24$
- $a_{12} = 2$, $M_{12} = \det \begin{bmatrix} 0 & 5 \\ 1 & 6 \end{bmatrix} = -5$, sign $-$ $\Rightarrow +10$
- $a_{13} = 3$, $M_{13} = \det \begin{bmatrix} 0 & 4 \\ 1 & 0 \end{bmatrix} = -4$, sign $+$ $\Rightarrow -12$

$$\det(A) = 24 + 10 - 12 = \mathbf{22}$$

2.10 The Inverse of a Matrix

Undoing a Transformation

Definition

A^{-1} is the matrix that undoes A :

$$AA^{-1} = A^{-1}A = I$$

For 2×2 there is a closed form:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Swap the diagonal, negate the off-diagonal,
divide by det.

Numerical Example

$$A = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}, \quad \det(A) = 24 - 14 = 10$$

$$A^{-1} = \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}$$

2.10 When the Inverse Fails

Singularity, and the Rules of Inversion

The Singular Case

If $\det(A) = 0$ the 2×2 formula divides by zero. The matrix is **singular**: it has no inverse.

Geometrically it collapsed space onto a line or a point. Information was destroyed, so there is nothing to undo it with — the same many-to-one idea from Part I.

Three Ways to Say the Same Thing

For a square A , these are all equivalent:

$\det(A) = 0 \iff \text{rank} < n \iff \lambda = 0 \text{ for some } i$
and any one of them means **no inverse**.

Rules Worth Memorising

- $(AB)^{-1} = B^{-1}A^{-1}$ — order reverses
- $(A^T)^{-1} = (A^{-1})^T$
- $(A^{-1})^{-1} = A$
- $\det(A^{-1}) = 1/\det(A)$

The reversal rule is intuitive: to undo “socks then shoes”, remove shoes first.

Machine Learning Connection

A near-singular matrix is almost as bad as a singular one: the inverse becomes enormous and numerically unstable. This is what “**ill-conditioned**” means in practice.

2.10 Inverse of a 3×3 : The Algorithm

Determinant, Cofactors, Adjugate

The Recipe

$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$, where $\text{adj}(A) = C^T$ is the **transpose of the cofactor matrix**.

Step 1 — Determinant

For $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, along row 1:

$$\det(A) = 1 \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} - 0 + 1 \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1$$

Non-zero, so the inverse exists.

Step 2 — Cofactor Matrix

$$C = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & -1 \\ -1 & -1 & 1 \end{bmatrix}$$

E.g. $C_{12} = - \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = 1$.

Steps 3 & 4 — Adjugate, then Divide

Here C is symmetric, so $\text{adj}(A) = C^T = C$:

$$A^{-1} = \frac{1}{-1} C = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

2.11 Solving a System of Linear Equations

What Inverses Are Actually For

Systems Become $A\mathbf{x} = \mathbf{b}$

$$\begin{array}{l} 2x + y = 5 \\ x + 3y = 10 \end{array} \iff \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

If A^{-1} exists, the solution is immediate:

$$\mathbf{x} = A^{-1}\mathbf{b}$$

Numerical Example

$\det(A) = 6 - 1 = 5$, so

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}, \quad \mathbf{x} = \frac{1}{5} \begin{bmatrix} 5 \\ 15 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Check: $2(1) + 3 = 5 \checkmark$ and $1 + 3(3) = 10 \checkmark$

Three Possible Outcomes

- $\det \neq 0$: exactly **one** solution (lines cross once).
- $\det = 0$, consistent: **infinitely many** (same line twice).
- $\det = 0$, inconsistent: **no** solution (parallel lines).

Machine Learning Connection

In practice nobody computes A^{-1} to solve a system — it is slow and numerically fragile. Libraries use **Gaussian elimination** or matrix factorisations instead. `numpy.linalg.solve` is the right tool, not `inv`.

2.12 Eigenvalues and Eigenvectors

The Directions a Matrix Does Not Turn

Definition

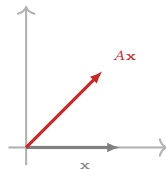
For a square A , an **eigenvector** $\mathbf{v}$ is a non-zero vector whose direction survives the transformation — only its length changes:

$$A\mathbf{v} = \lambda\mathbf{v}$$

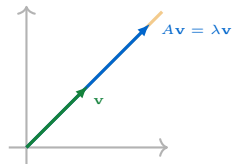
- $\mathbf{v}$ — the eigenvector (a direction)
- λ — the eigenvalue (the stretch factor)

Reading λ

$|\lambda| > 1$ stretches, $|\lambda| < 1$ shrinks, $\lambda = 1$ leaves it untouched, and $\lambda < 0$ flips it end-to-end.



not an eigenvector
direction turned



an eigenvector
same line, stretched

2.13 Finding Eigenvalues

The Characteristic Equation

Where It Comes From

Rearranging $A\mathbf{v} = \lambda\mathbf{v}$ gives $(A - \lambda I)\mathbf{v} = \mathbf{0}$.

For a **non-zero** $\mathbf{v}$ to exist, $A - \lambda I$ must squash something to zero — that is, be singular:

$$\det(A - \lambda I) = 0$$

This is the **characteristic equation**.

A Free Shortcut

For a **triangular** (or diagonal) matrix, the eigenvalues are simply the **diagonal entries** — no algebra needed.

Numerical Example

For $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ (triangular): $\lambda = 2, 2, 3$.

Eigenvector for $\lambda = 3$: solve $(A - 3I)\mathbf{v} = \mathbf{0}$:

$$\begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0}$$

Row 2 gives $y = 0$; row 1 then gives $x = 0$; z is free. Taking $z = 1$:

$$\mathbf{v}_3 = [0, 0, 1]^T$$

2.13 Eigenvalues: A Fuller Example

Working Through a 3×3

The Matrix

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{bmatrix}$$

Expanding $\det(A - \lambda I) = 0$ along the first column:

$$(2 - \lambda)[(3 - \lambda)(9 - \lambda) - 16] = 0$$

The bracket simplifies to
 $\lambda^2 - 12\lambda + 11 = (\lambda - 1)(\lambda - 11)$, giving

$$\lambda_1 = 2, \quad \lambda_2 = 1, \quad \lambda_3 = 11$$

Numerical Example

Eigenvector for $\lambda = 2$:

$$(A - 2I) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 4 & 7 \end{bmatrix}$$

Rows 2 and 3 give $y + 4z = 0$ and $4y + 7z = 0$; together these force $y = z = 0$, while x is free.

$$\mathbf{v}_1 = [1, 0, 0]^T$$

Eigenvectors are only defined up to scale — any non-zero multiple works, so we usually report the unit-length one.

2.14 Properties of Eigenvalues

Two Free Sanity Checks

Trace and Determinant

For an $n \times n$ matrix:

$$\sum_{i=1}^n \lambda_i = \text{tr}(A) \quad \prod_{i=1}^n \lambda_i = \det(A)$$

where the **trace** $\text{tr}(A)$ is the sum of the diagonal entries.

Numerical Example

For $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$, the eigenvalues are 5 and 2.

- $\text{tr}(A) = 4 + 3 = 7 = 5 + 2 \checkmark$
- $\det(A) = 12 - 2 = 10 = 5 \times 2 \checkmark$

An Important Consequence

Since the eigenvalues multiply to give $\det(A)$:

$$\lambda = 0 \text{ for some } i \iff \det(A) = 0$$

So a **zero eigenvalue means a singular matrix** — no inverse, rank-deficient, information lost.

Machine Learning Connection

Eigenvalues **rank a matrix's directions** by importance. A large $|\lambda|$ marks a direction it acts on strongly; a near-zero one, a direction it almost flattens; a zero one, a direction it destroys outright.

2.15 Symmetric Matrices Are Special

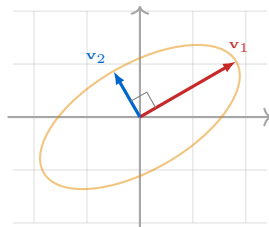
Why We Care So Much About $A = A^T$

The Spectral Theorem

Every **real symmetric** matrix is guaranteed to have:

- 1 eigenvalues that are all **real** (never complex), and
- 2 eigenvectors that are mutually **orthogonal**.

A symmetric matrix always supplies a clean, right-angled set of axes.



Why This Is Lucky

The two matrices ML cares about most — the **covariance matrix** and the **Hessian** — are both symmetric, so their eigen-analysis is always well behaved.

The eigenvectors of a symmetric matrix meet at exactly 90° — they form the natural axes of the ellipse it describes.

2.16 Positive Definiteness

Reading Curvature Off the Eigenvalues

Classification by Sign

A symmetric matrix is called

- **positive definite** if all $\lambda_i > 0$
- **negative definite** if all $\lambda_i < 0$
- **indefinite** if the signs are mixed
- **semi-definite** if some $\lambda_i = 0$

The Intuition

Eigenvalues are stretch factors along the matrix's own natural axes. All positive $\Rightarrow$ the surface **curves up in every direction** (a bowl); all negative $\Rightarrow$ a dome; mixed $\Rightarrow$ up one way, down another.

Numerical Example

$$H = \begin{bmatrix} 6 & 0 \\ 0 & 8 \end{bmatrix}$$

Diagonal, so $\lambda = 6, 8$. Both positive $\Rightarrow$ **positive definite** $\Rightarrow$ a bowl.

$$H = \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$$

$\lambda = 2, -3$. Mixed signs $\Rightarrow$ **indefinite** $\Rightarrow$ a saddle.

2.16 Keeping the Part I Promise

The General Second-Derivative Test

What We Postponed

In Part I we found critical points with $\nabla f = 0$, but could only classify them using the 2-variable D -test or a diagonal Hessian — because the general test needs eigenvalues.

We now have eigenvalues.

The Full Test

At a critical point $\mathbf{x}_0$ where $\nabla f(\mathbf{x}_0) = \mathbf{0}$, compute the eigenvalues of the Hessian H there:

Eigenvalues of H	Verdict
all $\lambda_i > 0$	Local minimum
all $\lambda_i < 0$	Local maximum
mixed signs	Saddle point
some $\lambda_i = 0$	Inconclusive

Machine Learning Connection

A positive-definite Hessian everywhere means the loss is **convex**: one global minimum, and gradient descent cannot get stuck. Deep networks are **not** convex — which is why saddle points matter.

2.17 Worked Example: The Second-Derivative Test

Step 1 — Find the Critical Points

The Function

$$f(x, y) = x^3 + y^3 - 3xy$$

Set the gradient to zero:

$$\nabla f = (3x^2 - 3y, 3y^2 - 3x) = \vec{0}$$

which gives $y = x^2$ and $x = y^2$.

Solving the Pair

Substituting, $x = (x^2)^2 = x^4$, so $x(x^3 - 1) = 0$, giving $x = 0$ or $x = 1$.

Critical points: $(0, 0)$ and $(1, 1)$. The gradient alone cannot say what kind they are.

Step 2 — Build the Hessian

The second derivatives are $f_{xx} = 6x$, $f_{yy} = 6y$ and $f_{xy} = f_{yx} = -3$:

$$H(x, y) = \begin{bmatrix} 6x & -3 \\ -3 & 6y \end{bmatrix}$$

Symmetric, as a Hessian always is — so its eigenvalues are guaranteed real.

Note the Difficulty

H is **not diagonal** here, so its type cannot be read off by eye. This is exactly the case Part I had to postpone until we had eigenvalues.

2.17 Worked Example: Classifying Both Points

Step 3 — Eigenvalues of the Hessian

Numerical Example

At (0,0): $H = \begin{bmatrix} 0 & -3 \\ -3 & 0 \end{bmatrix}$

$$\det(H - \lambda I) = \lambda^2 - 9 = 0 \Rightarrow \lambda = +3, -3$$

Mixed signs $\Rightarrow$ indefinite $\Rightarrow$ **saddle point**.

At (1,1): $H = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$

$$(6 - \lambda)^2 - 9 = 0 \Rightarrow \lambda = 3, 9$$

Both positive $\Rightarrow$ positive definite $\Rightarrow$ **local minimum**, with $f(1,1) = -1$.

A Free Check, Using Slide 2.14

	at (0,0)	at (1,1)
$\text{tr}(H)$	0	12
$\sum \lambda_i$	$3 - 3 = 0 \checkmark$	$3 + 9 = 12 \checkmark$
$\det(H)$	-9	27
$\prod \lambda_i$	$3(-3) = -9 \checkmark$	$3 \times 9 = 27 \checkmark$

Machine Learning Connection

For a 2×2 Hessian a **negative determinant** is an instant give-away: the eigenvalues must then have opposite signs, so the point is a saddle — no eigenvalue computation needed.

Chapter 2 Take-Away: Matrices

- 1 A **matrix** is a grid of numbers — and a **function** that rotates, stretches, shears or squashes vectors.
- 2 Ax is a **linear combination of A 's columns**, so the output can never escape their span.
- 3 **Shapes must match:** $(m \times n)(n \times p) = (m \times p)$. And $AB \neq BA$ — you may regroup, never reorder.
- 4 **Rank** counts genuinely independent directions. Low rank means redundant features.
- 5 $\det = 0$, **rank** $< n$, and $\lambda = 0$ **all say the same thing:** the matrix is singular and cannot be inverted.
- 6 **Eigenvectors** are the directions a matrix does not turn; **eigenvalues** say by how much it stretches them.
- 7 **All eigenvalues positive $\Rightarrow$ a bowl.** That is the general second-derivative test, and the definition of a convex loss.

Chapter 2 Quiz

Test Your Skills

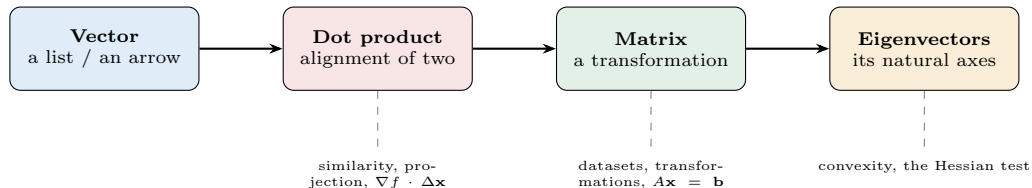
- ❶ Can you multiply a 4×2 matrix by another 4×2 matrix?
- ❷ If A is symmetric, what is $A - A^T$?
- ❸ If $\det(A) = 5$ for a 2×2 matrix, what is $\det(2A)$?
- ❹ If a matrix has a column of all zeros, what is its determinant?
- ❺ If $\lambda = 0$ is an eigenvalue, is the matrix invertible?
- ❻ For $A = \begin{bmatrix} 1 & k \\ 2 & 4 \end{bmatrix}$, what value of k makes it singular?
- ❼ True or false: $(AB)^{-1} = A^{-1}B^{-1}$.
- ❽ True or false: $(AB)^T = B^T A^T$.
- ❾ A 3×3 symmetric matrix has eigenvalues 4, 1, -2 . Minimum, maximum or saddle?

Answers are in the **Appendix** at the end of the deck.

Wrapping Up

The Whole Deck on One Slide

How the Ideas Connect



The Sentence to Walk Away With

A vector is a point of data; a matrix is something you do to it; its eigenvectors tell you what it really does.

Where This Goes Next

Part III: Optimization With Constraints

You Now Have the Full Toolkit

Parts I and II together cover unconstrained optimization end to end: slopes, gradients, curvature, and the eigenvalue test that classifies any critical point.

Coming Up

- Constrained optimization — “minimise the loss, *but* keep the weights small”
- **Lagrange multipliers**
- Singular Value Decomposition (SVD), the non-square cousin of eigen-decomposition
- Matrix calculus: derivatives with respect to vectors and matrices

Machine Learning Connection

Almost every regularised model is a constrained optimization in disguise: “minimise the error *subject to* a bound on $\|\mathbf{w}\|$ ”. That is why the norms from Chapter 1 keep reappearing.

Same landscape, same tools — now with fences on it.

Thank You

Questions?

Appendix: Quiz Answers

Answers — Chapter 1

Vectors

- ❶ $\|\mathbf{u}\| = \sqrt{4+4} = 2\sqrt{2}$, so $\hat{\mathbf{u}} = \left[\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right]$.
- ❷ $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos 90^\circ = 0$.
- ❸ $|-5| + |10| + |-2| = \mathbf{17}$.
- ❹ **False.** $\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2$, a sum of squares, so it can never be negative.
- ❺ $\mathbf{y} = 2\mathbf{x}$, so they are collinear and $\cos \theta = \mathbf{1}$.
- ❻ The vectors are **orthogonal** (perpendicular).
- ❼ **No** — they are dependent, since $[-2, -6] = -2[1, 3]$.
- ❽ Only the x -**axis** — a line, not the plane. The second vector adds no new direction.

Answers — Chapter 2

Matrices

- ❶ **No.** The inner dimensions (2 and 4) do not match.
- ❷ The **zero matrix**, since $A = A^T$ by definition of symmetric.
- ❸ **20.** For an $n \times n$ matrix $\det(cA) = c^n \det(A)$, so $2^2 \times 5 = 20$.
- ❹ **0.** A zero column makes the columns linearly dependent, so the matrix is singular.
- ❺ **No.** The eigenvalues multiply to $\det(A)$, so $\lambda = 0$ forces $\det(A) = 0$.
- ❻ Singular when $\det = 4 - 2k = 0$, so $\mathbf{k} = \mathbf{2}$.
- ❼ **False.** The order reverses: $(AB)^{-1} = B^{-1}A^{-1}$.
- ❽ **True.** $(AB)^T = B^T A^T$ — the order reverses here too.
- ❾ A **saddle point**, because the eigenvalues have mixed signs.